

# VANSH TYAGI

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## EDUCATION

### National Institute of Technology Kurukshetra

B.Tech in Electronics and Communication Engineering; CGPA: 9.100/10.0

Kurukshetra, India

Aug 2024 – Expected May 2028

## PROJECTS

### Full-Stack Blog Platform (MegaBlog) | [GitHub](#)

Jul 2026 – Present

- Developed a scalable, component-driven full-stack blogging platform using React, Vite, Tailwind CSS, and Appwrite.
- Engineered secure user authentication and session management using Appwrite, seamlessly integrated with Redux Toolkit for global state synchronization.
- Implemented protected routing via React Router and built highly reusable, validated UI components using React Hook Form.
- Architected a modular codebase adhering strictly to separation of concerns for services, configuration, and UI.
- Currently extending the platform with blog management features, content creation workflows, and backend integration.

### Smart Delivery Route Optimizer | [GitHub](#)

June 2026

- Engineered a dynamic routing simulation in Python, bridging Machine Learning predictions with fundamental Graph algorithms.
- Developed a K-Nearest Neighbors (KNN) regressor to predict real-time traffic multipliers based on variables like time and weather severity.
- Implemented Dijkstra's Algorithm utilizing a Min-Heap (Priority Queue) to dynamically compute the absolute shortest path on a weighted adjacency list.

## TECHNICAL SKILLS

**Languages:** C/C++, Python, JavaScript, SQL

**Web Development:** HTML5, CSS3, Tailwind CSS, React, React Router, Redux Toolkit, React Hook Form, Node.js

**Machine Learning:** scikit-learn

**Data Science:** NumPy, pandas, Matplotlib, Seaborn, Jupyter

**Databases:** MySQL, Appwrite

**Tools & Systems:** Git, Linux, VS Code, Vite

## ACHIEVEMENTS

### JEE Main 2024 | 99.36 Percentile

Jan 2024

- Secured the 99.36 percentile among 1.2 million candidates in one of India's most competitive engineering entrance examinations.

### Smart India Hackathon | Participant

2026

- Participated in India's largest hackathon initiative, collaborating on innovative tech solutions for real-world problems.